

PART 1

$$P = (0, 0) \quad Q = (x, x^2 + 5)$$

$$d(x) = \sqrt{(x - x_0)^2 + (y - y_0)^2} = \sqrt{(x - 0)^2 + (x^2 + 5 - 0)^2} \\ = \sqrt{x^2 + (x^2 + 5)^2}$$

$$D(x) = d(x)^2 = x^2 + (x^2 + 5)^2 \\ = x^2 + x^4 + 10x^2 + 25 \\ = x^4 + 11x^2 + 25$$

$$D'(x) = 4x^3 + 22x = 0$$

$$2x(2x^2 + 11) = 0 \quad 2x^2 + 11 > 0 \text{ for every real } x$$

$$\therefore \underline{x = 0}$$

$$D''(x) = 12x^2 + 22 > 0$$

$\therefore 12x^2 + 22 > 0$ for $x \in \mathbb{R}$, it's point is the unique global minimum

$$\therefore x = 0, \therefore y = (0)^2 + 5 = 5$$

closest point: $(0, 5)$

$$d_{\min} = \sqrt{(0-0)^2 + (5-0)^2}$$

$$d_{\min} = 5$$

$$P = (-4, 0) \quad Q = (x, x^2 + 5)$$

$$D(x) = (x + 4)^2 + (x^2 + 5)^2$$

$$= x^2 + 8x + 16 + x^4 + 10x^2 + 25$$

$$= x^4 + 11x^2 + 8x + 41$$

$$D'(x) = 4x^3 + 22x + 8 = 0$$

$$x^3 + \frac{11}{2}x + 2 = 0$$

for $x^3 + px + q = 0$:

$$x = \sqrt[3]{-\frac{q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}}$$

$$x^* = \sqrt[3]{-1 + \sqrt{\frac{1547}{216}}} + \sqrt[3]{-1 - \sqrt{\frac{1547}{216}}}$$

$$= -0.355$$

closest point: $(-0.355, 5.126)$

$$D''(x) = 3x^2 + \frac{11}{2} > 0 \quad \therefore \text{unique global min}$$

$$d_{\min} = 6.2898$$

$$y^* = (-0.355)^2 + 5 = 5.126$$

$$P = (-8, 0) \quad Q = (x, x^2 + 5)$$

$$\begin{aligned} D(x) &= (x+8)^2 + (x^2+5)^2 \\ &= x^2 + 16x + 64 + x^4 + 10x^2 + 25 \\ &= x^4 + 11x^2 + 16x + 89 \end{aligned}$$

$$\begin{aligned} D'(x) &= 4x^3 + 22x + 16 \\ &= x^3 + \frac{11}{2}x + 4 = 0 \quad x = -0.672 \end{aligned}$$

$$D''(x) = 3x^2 + \frac{11}{2} > 0 \text{ for } x \in \mathbb{R} \quad \therefore \text{unique global min}$$

$$\begin{aligned} y &= (-0.672)^2 + 5 \\ &= 5.4517 \end{aligned} \quad \begin{aligned} &\text{closest point: } (-0.672, 5.4517) \\ &d_{\min} = 9.1334 \end{aligned}$$

$$P = (2, 0) \quad Q = (x, x^2 + 5)$$

$$\begin{aligned} D(x) &= (x-2)^2 + (x^2+5)^2 \\ &= x^2 - 4x + 4 + x^4 + 10x^2 + 25 \\ &= x^4 + 11x^2 - 4x + 29 \end{aligned}$$

$$\begin{aligned} D'(x) &= 4x^3 + 22x - 4 \\ &= x^3 + \frac{11}{2}x - 1 = 0 \\ x &= 0.1807 \end{aligned}$$

$$D''(x) = 3x^2 + \frac{11}{2} > 0 \text{ for } x \in \mathbb{R} \quad \therefore \text{unique global min}$$

$$\begin{aligned} y &= 5.03 \\ &\text{closest point: } (0.1807, 5.03) \\ &d_{\min} = 5.35 \end{aligned}$$

$$P = (6, 0) \quad Q = (x, x^2 + 5)$$

$$\begin{aligned} D(x) &= (x-6)^2 + (x^2+5)^2 \\ &= x^4 + 11x^2 - 12x + 61 \end{aligned}$$

$$\begin{aligned} D'(x) &= 4x^3 + 22x - 12 = 0 \\ x^3 + \frac{11}{2}x - 3 &= 0 \\ x &= 0.5199 \end{aligned}$$

$$D''(x) = 3x^2 + \frac{11}{2} > 0 \text{ for } x \in \mathbb{R} \therefore \text{unique global min}$$

$$y = 5.27$$

$$\text{closest point: } (0.5199, 5.27)$$

$$d_{\min} = 7.603$$

Algorithm A: Newton-Raphson

Newton-Raphson Method:

$$D'(x) = 0$$

$$\text{let } g(x) = D'(x)$$

$$x_{n+1} = x_n - \frac{g(x_n)}{g'(x_n)} \Rightarrow x_{n+1} = x_n - \frac{D'(x_n)}{D''(x_n)}$$

$$D(x) = (x-x_0)^2 + (f(x)-y_0)^2$$

$$D'(x) = 2(x-x_0) + 2(f(x)-y_0)f'(x)$$

$$D''(x) = 2 + 2[f'(x)]^2 + 2(f(x)-y_0)f''(x)$$

for $f(x) = x^2 + 5$, $f'(x) = 2x$, $f''(x) = 2$ and All $y_0 = 0$:

$$\begin{aligned} D'(x) &= 2(x-x_0) + 2(x^2+5)(2x) \\ &= 4x^3 + 2x + 20x - 2x_0 \\ &= 4x^3 + 22x - 2x_0 \end{aligned}$$

$$\begin{aligned} D''(x) &= 2 + 2(2x)^2 + 2(x^2+5)(2) \\ &= 2 + 8x^2 + 4x^2 + 20 \\ &= 12x^2 + 22 \end{aligned}$$

$$\therefore x_{n+1} = x_n - \frac{4x_n^3 + 22x_n - 2x_0}{12x_n^2 + 22}$$

$$P=(0,0) \quad Q=(x, x^2+5)$$

$$x_{n+1} = x_n - \frac{4x_n^3 + 22x_n}{12x_n^2 + 22}$$

$$x_3 = \frac{4}{17} - \frac{4(0.235)^3 + 22(0.235)}{12(0.235)^2 + 22}$$

$$= 4.598 \cdot 10^{-3}$$

$$x_4 = x_3 - \frac{4x_3^3 + 22x_3}{12x_3^2 + 22}$$

$$x_4 = 3.53 \cdot 10^{-8} \approx 0$$

$$\therefore x^* \approx 0$$

$$y^* = x^2 + 5 = 5$$

closest point: (0, 5)
 $d^* = 5$

$$P=(-4,0) \quad Q=(x, x^2+5)$$

$$x_{n+1} = x_n - \frac{4x_n^3 + 22x_n - 2x_0}{12x_n^2 + 22}$$

$$x_{n+1} = x_n - \frac{4x_n^3 + 22x_n + 8}{12x_n^2 + 22}$$

$$x_3 = 0 - \frac{8}{22}$$

$$x_3 = -\frac{4}{11} = -0.36$$

closest point: (-0.355, 5.126)
 $d^* = 6.6898$

$$\text{assume } x_1 = 1$$

$$x_2 = 1 - \frac{4(1)^3 + 22(1)}{12(1)^2 + 22}$$

$$= 1 - \frac{26}{34}$$

$$= \frac{4}{17} \approx 0.235$$

$$\text{assume } x_1 = 1$$

$$x_2 = 1 - \frac{34}{34}$$

$$x_2 = 0$$

$$x_4 = x_3 - \frac{4(x_3)^3 + 22(x_3) + 8}{12(x_3)^2 + 22}$$

$$x_4 = -0.355$$

$$x^* = -0.355$$

$$y^* = 5.126$$

$$P = (-8, 0) \quad Q = (x, x^2 + 5)$$

$$x_{n+1} = x_n - \frac{4x_n^3 + 22x_n - 2x_0}{12x_n^2 + 22}$$

$$x_{n+1} = x_n - \frac{4x_n^3 + 22x_n + 16}{12x_n^2 + 22}$$

$$x_3 = x_2 - \frac{4(x_2)^3 + 22(x_2) + 16}{12(x_2)^2 + 22}$$

$$x_3 = -0.71$$

assume $x_1 = 1$

$$x_2 = 1 - \frac{42}{34}$$

$$x_2 = -\frac{4}{17} \approx -0.235$$

$$x_4 = x_3 - \frac{4(x_3)^3 + 22(x_3) + 16}{12(x_3)^2 + 22}$$

$$x_4 = -0.67$$

$$x^* = -0.67$$

$$y^* = 5.45$$

closest point: $(-0.67, 5.45)$

$$d^* = 9.13$$

$$P = (2, 0) \quad Q = (x, x^2 + 5)$$

$$x_{n+1} = x_n - \frac{4x_n^3 + 22x_n - 2x_0}{12x_n^2 + 22}$$

$$x_{n+1} = x_n - \frac{4x_n^3 + 22x_n - 4}{12x_n^2 + 22}$$

$$\text{let } x_1 = 1$$

$$x_2 = (1) - \frac{4(1)^3 + 22(1) - 4}{12(1)^2 + 22}$$

$$x_2 = 1 - \frac{22}{34} = \frac{6}{17} \approx 0.35$$

$$x_3 = x_2 - \frac{4x_2^3 + 22x_2 - 4}{12(x_2)^2 + 22}$$

$$x_3 = 0.185$$

$$x_4 = x_3 - \frac{4x_3^3 + 22x_3 - 4}{12x_3^2 + 22}$$

$$x_4 = 0.1807$$

$$x^* = 0.180746$$

$$y^* = 5.03267$$

closest point: $(0.1807, 5.03267)$

$$d^* = 5.35$$

$$P = (6, 0) \quad Q = (x, x^2 + 5)$$

assume $x_1 = 1$

$$x_{n+1} = x_n - \frac{4x_n^3 + 22x_n - 2x_0}{12x_n^2 + 22}$$

$$x_2 = 1 - \frac{14}{34}$$

$$x_{n+1} = x_n - \frac{4x_n^3 + 22x_n - 12}{12x_n^2 + 22}$$

$$x_2 = \frac{10}{17} \approx 0.588$$

$$x_3 = x_2 - \frac{4x_2^3 + 22x_2 - 12}{12x_2^2 + 22}$$

$$x_4 = x_3 - \frac{4x_3^3 + 22x_3 - 12}{12x_3^2 + 22}$$

$$x_3 = 0.5199$$

$$x_4 = 0.5199$$

$$x^* = 0.5199$$

$$y^* = 5.2703$$

closest point: $(0.5199, 5.2703)$

$$d^* = 7.603$$

Algorithm B: Golden Section Search

$$f(x) = x^2 + 5$$

$$D(x) = (x - x_0)^2 + (f(x) - y_0)^2 \quad \text{since } y_0 = 0$$

$$= (x - x_0)^2 + (x^2 + 5)^2$$

$$\phi = \frac{1 + \sqrt{5}}{2} \approx 1.618$$

$$r = 2 - \phi$$

$$r \approx 0.382$$

$$x_1 = a + r(b-a) \quad x_2 = b - r(b-a)$$

$$P = (0, 0)$$

$$D(x) = x^2 + (x^2 + 5)^2$$

$$\text{use } a = -2, b = 2$$

$$[-2, 2]$$

$$x_1 = a + c(b-a) \quad x_2 = b - c(b-a)$$

$$x_1 = -2 + 4c \quad x_2 = 2 - 4c$$

$$x_1 \approx -0.472 \quad x_2 \approx 0.472$$

$$D(-0.472) = (-0.472)^2 + [(-0.472)^2 + 5]^2$$

$$D(-0.472) \approx 27.51 = D(0.472) \approx 27.51$$

$$\therefore a = x_1 = -0.472$$

$$[-0.472, 2]$$

$$\begin{aligned} x_1 &= 0.472 & x_2 &= b - c(b-a) \\ & & &= 2 - c(2 - (-0.472)) \\ & & &= 1.056 \end{aligned}$$

$$D(0.472) = 27.51 < D(1.056) = (1.056)^2 + (1.056^2 + 5)^2 = 38.5$$

$$\therefore b = x_2 = 1.056$$

$$[-0.472, 1.056]$$

$$\begin{aligned} x_1 &= a + c(b-a) & x_2 &= 0.472 \\ &= -0.472 + c(1.056 + 0.472) \\ &= 0.112 \end{aligned}$$

$$D(0.112) = 25.14 < D(0.472) = 27.51$$

$$\therefore b = x_2 = 0.472$$

$$[a, b] \approx [-0.472, 0.472]$$

$$P = (-4, 0)$$

$$D(x) = (x+4)^2 + (x^2+5)^2$$

$$[-2, 2]$$

$$x_1 = -0.472$$

$$x_2 = 0.472$$

$$D(-0.472) = (-0.472+4)^2 + ((-0.472)^2+5)^2 < D(0.472) = (0.472+4)^2 + ((-0.472)^2+5)^2$$

$$= 39.724 \qquad = 47.28$$

$$\therefore b = x_2 = 0.472$$

$$[-2, 0.472]$$

$$x_1 = a + r(b-a)$$

$$x_2 = b - r(b-a)$$

$$= -1.056$$

$$= -0.472$$

$$D(-1.056) = (-1.056+4)^2 + ((-1.056)^2+5)^2 > D(-0.472) = 39.724$$

$$= 46.06 \qquad \therefore a = x_1 = -1.056$$

$$[-1.056, 0.472]$$

$$x_1 = a + r(b-a)$$

$$x_2 = b - r(b-a)$$

$$= -0.472$$

$$= -0.112$$

$$D(-0.472) = 39.724 < D(-0.112) = (-0.112+4)^2 + ((-0.112)^2+5)^2$$

$$= 40.25$$

$$\therefore b = x_2 = -0.112$$

$$[a, b] \approx [-1.056, -0.112]$$

$$P = (-8, 0)$$

$$D(x) = (x+8)^2 + (x^2+5)^2$$

$$[-2, 2]$$

$$x_1 = a + r(b-a) \\ = -0.472$$

$$x_2 = b - r(b-a) \\ = 0.472$$

$$D(-0.472) = (-0.472+8)^2 + ((-0.472)^2+5)^2 < D(0.472) = (0.472+8)^2 + (0.472^2+5)^2 \\ = 84 \qquad \qquad \qquad = 99.1$$

$$\therefore b = x_2 = 0.472$$

$$[-2, 0.472]$$

$$x_1 = a + r(b-a) \\ = -1.056$$

$$x_2 = b - r(b-a) \\ = -0.472$$

$$D(-1.056) = (-1.056+8)^2 + ((-1.056)^2+5)^2 > D(-0.472) = 84 \\ = 85.6$$

$$\therefore a = x_1 = -1.056$$

$$[-1.056, 0.472]$$

$$x_1 = a + r(b-a) \\ = -0.472$$

$$x_2 = b - r(b-a) \\ = -0.112$$

$$D(-0.472) = 84.0 < D(-0.112) = (-0.112+8)^2 + ((-0.112)^2+5)^2 \\ = 87.3$$

$$\therefore b = x_2 = -0.112$$

$$[a, b] \approx [-0.472, -0.112]$$

$$P = (2, 0)$$

$$D(x) = (x-2)^2 + (x^2+5)^2$$

$$[-2, 2]$$

$$\begin{aligned} x_1 &= a + r(b-a) & x_2 &= b - r(b-a) \\ &= -0.472 & &= 0.472 \end{aligned}$$

$$\begin{aligned} D(-0.472) &= (-0.472-2)^2 + (-0.472^2+5)^2 > D(0.472) = (0.472-2)^2 + (0.472^2+5)^2 \\ &= 33.40 & &= 29.84 \end{aligned}$$

$$\therefore a = x_1 = -0.472$$

$$[-0.472, 2]$$

$$x_1 = 0.472 \quad x_2 = 1.056$$

$$\begin{aligned} D(0.472) &= 29.84 < D(1.056) = (1.056-2)^2 + (1.056^2+5)^2 \\ & & &= 38.50 \end{aligned}$$

$$\therefore b = x_2 = 1.056$$

$$[-0.472, 1.056]$$

$$x_1 = 0.112 \quad x_2 = 0.472$$

$$D(0.112) = 28.7 < D(0.472) = 29.84$$

$$\therefore b = x_2 = 0.472$$

$$[a, b] \approx [-0.472, 0.472]$$

$$P = (6, 0)$$

$$D(x) = (x-6)^2 + (x^2+5)^2$$

$$[-2, 2]$$

$$\begin{aligned} x_1 &= a + r(b-a) & x_2 &= b - r(b-a) \\ &= -0.472 & &= 0.472 \end{aligned}$$

$$\begin{aligned} D(-0.472) &= (-0.472-6)^2 + ((-0.472)^2+5)^2 > D(0.472) = (0.472-6)^2 + ((0.472)^2+5)^2 \\ &= 69.1 & &= 57.8 \end{aligned}$$

$$\therefore a = x_1 = -0.472$$

$$[-0.472, 2]$$

$$\begin{aligned} x_1 &= a + r(b-a) & x_2 &= b - r(b-a) \\ &= 0.472 & &= 1.056 \end{aligned}$$

$$\begin{aligned} D(0.472) = 57.8 < D(1.056) &= (1.056-6)^2 + (1.056^2+5)^2 \\ &= 62.1 \end{aligned}$$

$$\therefore b = x_2 = 1.056$$

$$[-0.472, 1.056]$$

$$\begin{aligned} x_1 &= a + r(b-a) & x_2 &= b - r(b-a) \\ &= 0.112 & &= 0.472 \end{aligned}$$

$$\begin{aligned} D(0.112) &= (0.112-6)^2 + (0.112^2+5)^2 > D(0.472) = 57.8 \\ &= 59.8 \end{aligned}$$

$$\therefore a = x_1 = 0.112$$

$$[a, b] \approx [0.112, 1.056]$$